

화학수학

다섯 번째 수업

다변수 미분 / 다변수 적분

Pfaffian Form: General Differential Form

(파피안형)

$$du = M(x, y)dx + N(x, y)dy$$

- Exact differential (완전 미분):

$$M(x, y) = \left(\frac{\partial u}{\partial x} \right)_y, \quad N(x, y) = \left(\frac{\partial u}{\partial y} \right)_x$$

- Inexact differential (불완전 미분): otherwise

Test of Exactness

$$du = M(x, y)dx + N(x, y)dy$$

Using the Euler reciprocity relation,

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$

if the differential is exact,

$$\left(\frac{\partial N}{\partial x} \right)_y = \left(\frac{\partial M}{\partial y} \right)_x$$

Integrating Factor

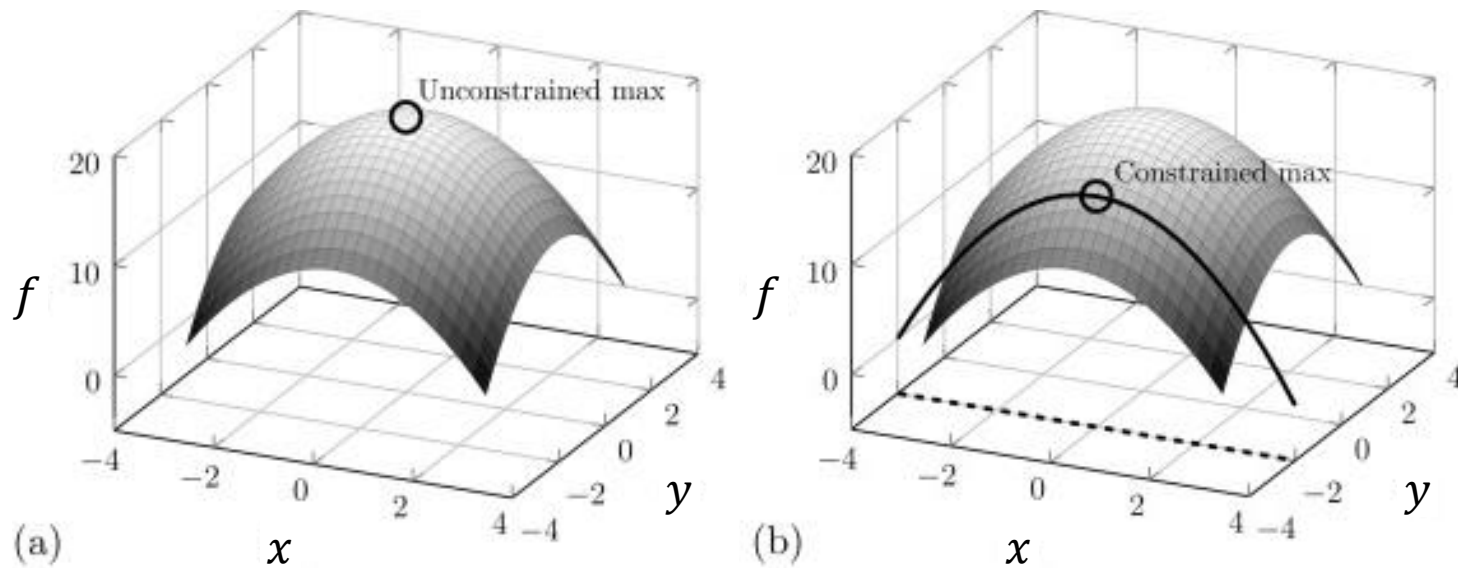
(적분인자)

inexact differential $\rightarrow$ exact differential
by multiplying a function (integrating factor)

No general method to find an integrating factor

Constrained Optimization

(제약 최적화)



with a constraint (제약 조건)

Lagrange's Method

Find the constrained extremum in $f(x, y)$,
when the constraint is written in the form $g(x, y) = 0$

1. New function $u(x, y) = f(x, y) + \lambda g(x, y)$
2. Partial derivatives of $u = 0$
3. Solve the equations from point 2 and $g(x, y) = 0$
to obtain x , y , and $\lambda \rightarrow$ constrained extrema

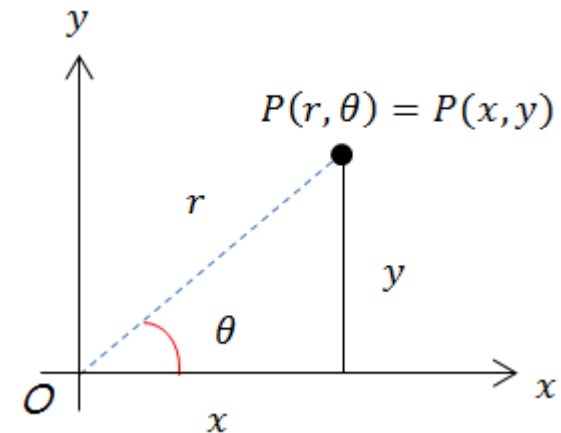
Coordinate Systems

- 2 dimensions

- Cartesian coordinates (직교 좌표)
- Polar coordinates (극좌표)

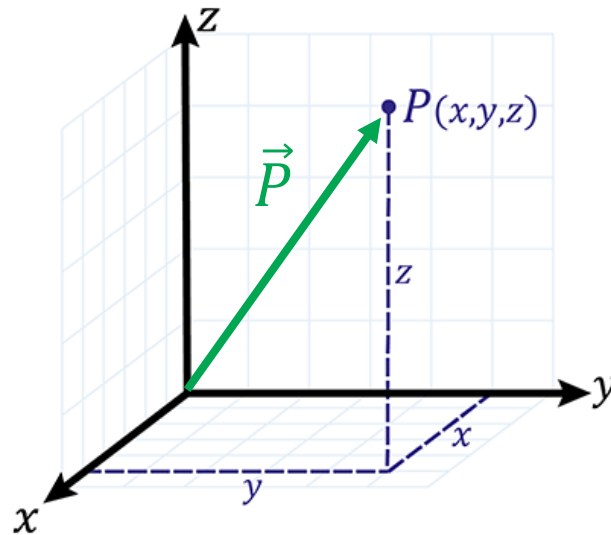
- 3 dimensions

- Cartesian coordinates (직교 좌표)
- Spherical (polar) coordinates (구면 좌표)
- Cylindrical (polar) coordinates (원통 좌표)



Vector

A quantity with magnitude and direction



(Image: <https://www.skillsyouneed.com/num/cartesian-coordinates.html>)

Scalar Product (내적)

The scalar product (dot product) of $\vec{A}$ and $\vec{B}$ is

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

and, equivalently,

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \alpha$$

where α is the angle between the two vectors.

Vector Product (외적)

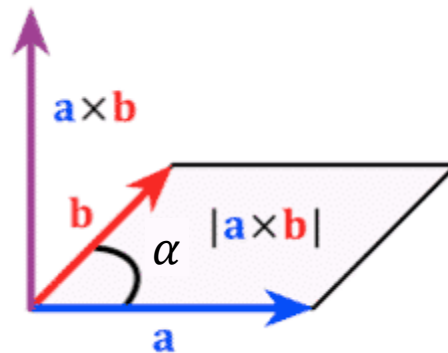
The vector product (cross product) of $\vec{A}$ and $\vec{B}$ is

$$\begin{aligned}\vec{A} \times \vec{B} &= \hat{i}(A_y B_z - A_z B_y) + \hat{j}(A_z B_x - A_x B_z) + \hat{k}(A_x B_y - A_y B_x) \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}\end{aligned}$$

Geometric Definition of Vector Product

For $\vec{C} = \vec{A} \times \vec{B}$,

- Magnitude: $|\vec{C}| = |\vec{A}||\vec{B}| \sin \alpha$ (α : angle between $\vec{A}$ & $\vec{B}$)
- Direction: right-hand rule



Today's Class

- Vector derivatives
 - Gradient, divergence, curl, Laplacian
 - In different coordinate systems
- Optimization
 - Constrained optimization
 - Lagrange's method

How to Differentiate Vectors?

- Scalar by scalar: $df/dx \rightarrow$ We know how to
- Scalar by vector: $df/d\vec{r}$
 $\rightarrow$ "Gradient"
- Vector by scalar: $d\vec{F}/dx$
 $\rightarrow$ Straightforward: $d\vec{F}/dx = (dF_x/dx, dF_y/dx, dF_z/dx)$
- Vector by vector: $d\vec{F}/d\vec{r}$
 $\rightarrow$ "Divergence" and "curl"

Vector Derivatives

- Gradient (기울기) ∇f

scalar $\rightarrow$ vector

- Divergence (발산) $\nabla \cdot \vec{F}$

vector $\rightarrow$ scalar

- Curl (회전) $\nabla \times \vec{F}$

vector $\rightarrow$ vector

- Laplacian (라플라스) $\nabla^2 f$

scalar $\rightarrow$ scalar (2nd order)

Gradient

$$\nabla f = \hat{i} \left(\frac{\partial f}{\partial x} \right) + \hat{j} \left(\frac{\partial f}{\partial y} \right) + \hat{k} \left(\frac{\partial f}{\partial z} \right) = \text{grad } f$$

$(\hat{i}, \hat{j}, \hat{k}$: unit vectors)

Example 8.18

Find the gradient of the function $f(x, y, z) = x^2 + y^2 + z^2$.

Using the definition of the gradient,

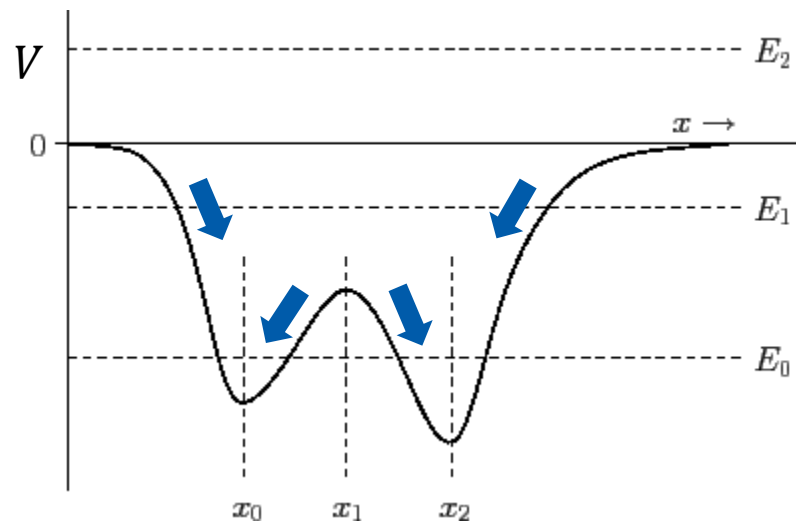
$$\nabla f = \hat{i} \left(\frac{\partial f}{\partial x} \right) + \hat{j} \left(\frac{\partial f}{\partial y} \right) + \hat{k} \left(\frac{\partial f}{\partial z} \right)$$

$$\nabla f = \hat{i}(2x) + \hat{j}(2y) + \hat{k}(2z) = 2(x\hat{i} + y\hat{j} + z\hat{k})$$

Example: Force and Potential

$$\vec{F} = -\nabla V$$

where $\vec{F}$ is the force and V is the potential energy



(Image: <http://farside.ph.utexas.edu/teaching/336k/Newtonhtml/node16.html>)

Example 8.19

Neglecting the attractions of all other celestial bodies, the gravitational potential energy of the earth and the sun is given by

$$V = -\frac{Gm_s m_e}{r},$$

where G is the universal gravitational constant, m_s and m_e are the masses of the sun and the earth, respectively, and r is the distance between the centers of the sun and the earth. Express the force on the earth.

Using the previous formula,

$$\vec{F} = -\nabla V = \nabla \left(\frac{Gm_s m_e}{r} \right) = Gm_s m_e \nabla \left(\frac{1}{r} \right) = Gm_s m_e \nabla \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right)$$

Example 8.19

$$\vec{F} = Gm_s m_e \nabla \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right)$$

$$\frac{\partial}{\partial x} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) = -\frac{1}{2} \frac{2x}{(x^2 + y^2 + z^2)^{3/2}} = -\frac{x}{r^3}$$

Similarly,

$$\frac{\partial}{\partial y} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) = -\frac{y}{r^3}, \quad \frac{\partial}{\partial z} \left(\frac{1}{\sqrt{x^2 + y^2 + z^2}} \right) = -\frac{z}{r^3}$$

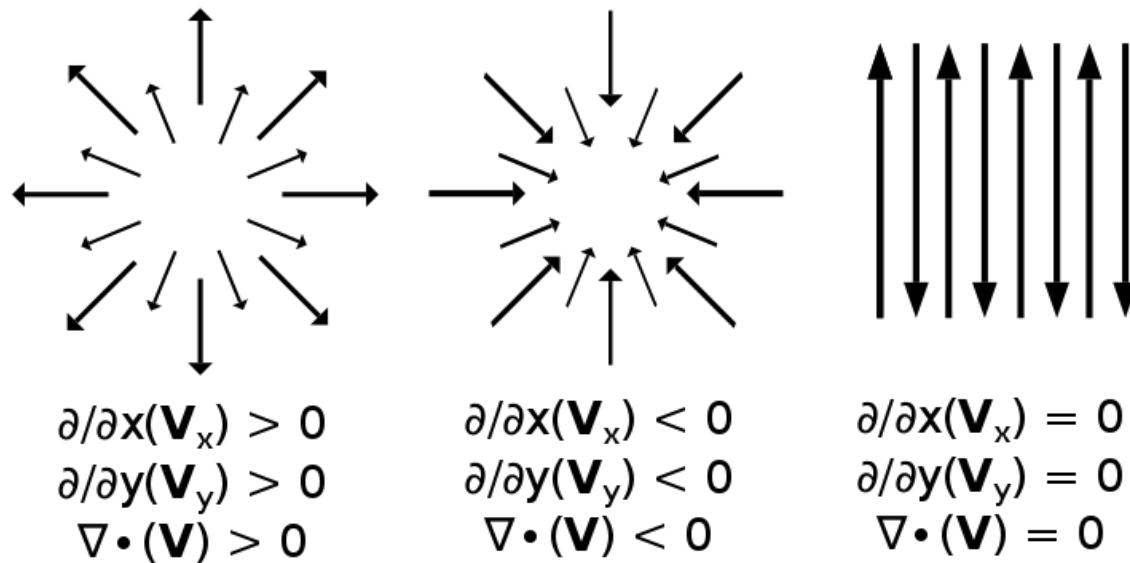
Hence,

$$\vec{F} = Gm_s m_e \left(-\frac{x}{r^3} \hat{i} - \frac{y}{r^3} \hat{j} - \frac{z}{r^3} \hat{k} \right) = -\frac{Gm_s m_e}{r^3} (x\hat{i} + y\hat{j} + z\hat{k})$$

Divergence

$$\nabla \cdot \vec{F} = \left(\frac{\partial F_x}{\partial x} \right) + \left(\frac{\partial F_y}{\partial y} \right) + \left(\frac{\partial F_z}{\partial z} \right) = \operatorname{div} \vec{F}$$

Meaning of Divergence



Example 8.20

Find $\nabla \cdot \vec{F}$ if $\vec{F}$ is given by

$$\vec{F} = \hat{i}x^2 + \hat{j}yz + \hat{k}\frac{xz^2}{y}.$$

Using the definition of the divergence,

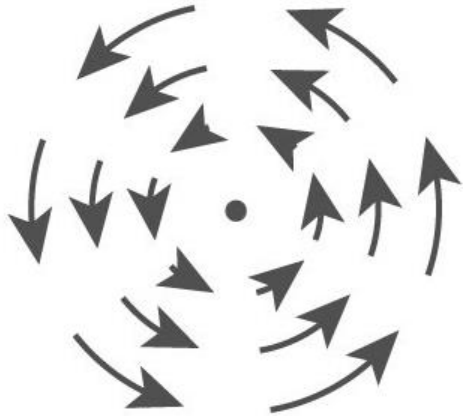
$$\nabla \cdot \vec{F} = \left(\frac{\partial F_x}{\partial x} \right) + \left(\frac{\partial F_y}{\partial y} \right) + \left(\frac{\partial F_z}{\partial z} \right)$$

$$\nabla \cdot \vec{F} = 2x + z + \frac{2xz}{y}$$

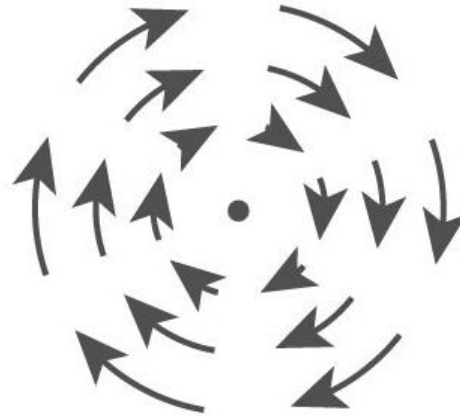
Curl

$$\begin{aligned}\nabla \times \vec{F} &= \hat{i} \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) + \hat{j} \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) + \hat{k} \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \\ &= \text{curl } \vec{F}\end{aligned}$$

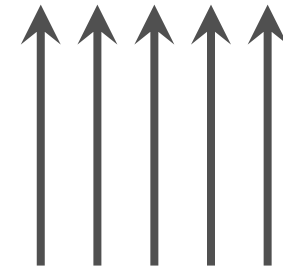
Meaning of Curl



$$(\nabla \times \vec{F})_z > 0$$



$$(\nabla \times \vec{F})_z < 0$$



$$(\nabla \times \vec{F})_z = 0$$

Example 8.21

Find $\nabla \times \vec{F}$ if $\vec{F}$ is $\vec{F} = \hat{i}y + \hat{j}z + \hat{k}x$.

Using the definition of the curl,

$$\nabla \times \vec{F} = \hat{i} \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) + \hat{j} \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) + \hat{k} \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right)$$

$$\nabla \times \vec{F} = \hat{i}(0 - 1) + \hat{j}(0 - 1) + \hat{k}(0 - 1) = -\hat{i} - \hat{j} - \hat{k}$$

Laplacian: divergence of gradient

$$\nabla^2 f = \left(\frac{\partial^2 f}{\partial x^2} \right) + \left(\frac{\partial^2 f}{\partial y^2} \right) + \left(\frac{\partial^2 f}{\partial z^2} \right)$$

Example 8.22

Find the Laplacian of the function

$$f(x, y, z) = A \sin(ax) \sin(by) \sin(cz).$$

Using the definition of the Laplacian,

$$\nabla^2 f = \left(\frac{\partial^2 f}{\partial x^2} \right) + \left(\frac{\partial^2 f}{\partial y^2} \right) + \left(\frac{\partial^2 f}{\partial z^2} \right)$$

$$(\partial^2 f / \partial x^2) = -Aa^2 \sin(ax) \sin(by) \sin(cz) = -a^2 f(x, y, z)$$

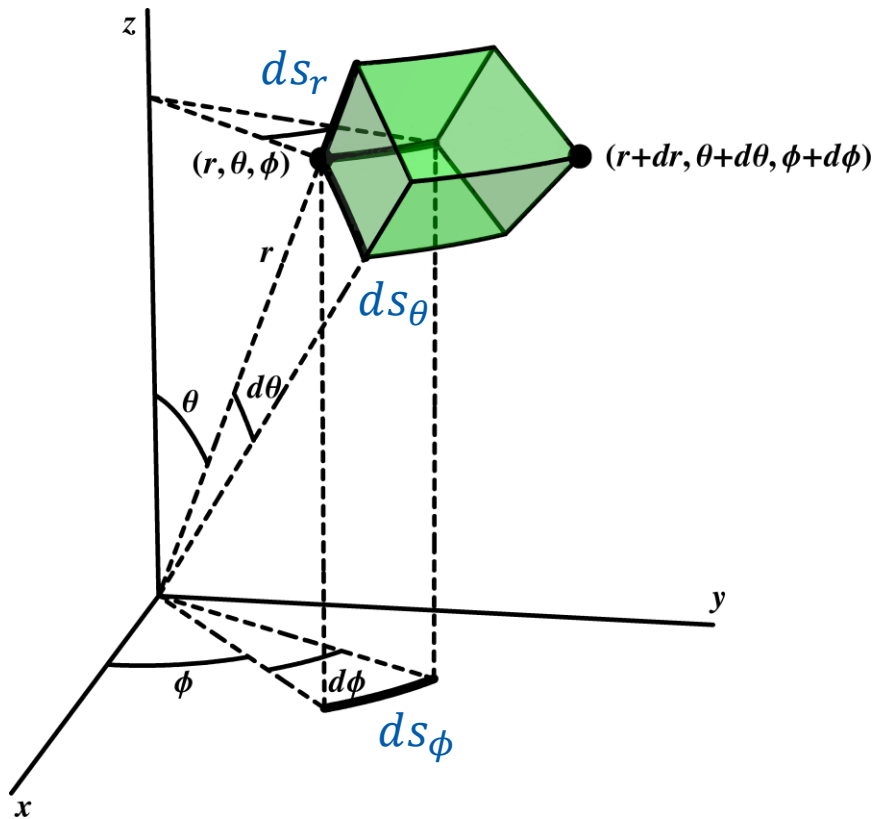
$$(\partial^2 f / \partial y^2) = -Ab^2 \sin(ax) \sin(by) \sin(cz) = -b^2 f(x, y, z)$$

$$(\partial^2 f / \partial z^2) = -Ac^2 \sin(ax) \sin(by) \sin(cz) = -c^2 f(x, y, z)$$

Hence,

$$\nabla^2 f = -(a^2 + b^2 + c^2)f(x, y, z)$$

Infinitesimal Displacements



$$ds_r = dr$$

$$ds_\theta = r d\theta$$

$$ds_\phi = r \sin \theta d\phi$$

Infinitesimal Displacements

In Cartesian coordinates,

$$d\vec{r} = \hat{i}dx + \hat{j}dy + \hat{k}dz$$

In spherical polar coordinates,

$$d\vec{r} = \hat{e}_r dr + \hat{e}_\theta r d\theta + \hat{e}_\phi r \sin \theta d\phi$$

In general,

$$d\vec{r} = \hat{e}_1 h_1 dq_1 + \hat{e}_2 h_2 dq_2 + \hat{e}_3 h_3 dq_3$$

Gradient

$$\nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$$

In the other coordinate system,

$$\nabla f = \hat{e}_1 \frac{1}{h_1} \frac{\partial f}{\partial q_1} + \hat{e}_2 \frac{1}{h_2} \frac{\partial f}{\partial q_2} + \hat{e}_3 \frac{1}{h_3} \frac{\partial f}{\partial q_3}$$

In spherical polar coordinates,

$$\nabla f = \hat{e}_r \frac{\partial f}{\partial r} + \hat{e}_\theta \frac{1}{r} \frac{\partial f}{\partial \theta} + \hat{e}_\phi \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi}$$

Divergence

$$\nabla \cdot \vec{F} = \left(\frac{\partial F_x}{\partial x} \right) + \left(\frac{\partial F_y}{\partial y} \right) + \left(\frac{\partial F_z}{\partial z} \right)$$

For $\vec{F} = \hat{e}_1 F_1 + \hat{e}_2 F_2 + \hat{e}_3 F_3$,

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial (F_1 h_2 h_3)}{\partial q_1} + \frac{\partial (F_2 h_1 h_3)}{\partial q_2} + \frac{\partial (F_3 h_1 h_2)}{\partial q_3} \right]$$

In spherical polar coordinates,

$$\nabla \cdot \vec{F} = \frac{1}{r^2 \sin \theta} \left[\frac{\partial (F_r r^2 \sin \theta)}{\partial r} + \frac{\partial (F_\theta r \sin \theta)}{\partial \theta} + \frac{\partial (F_\phi r)}{\partial \phi} \right]$$

Example 8.23(b)

Find the divergence of the position vector, which in spherical polar coordinates is $\vec{r} = \hat{e}_r r$.

Using the previous formula,

$$\nabla \cdot \vec{F} = \frac{1}{r^2 \sin \theta} \left[\frac{\partial(F_r r^2 \sin \theta)}{\partial r} + \frac{\partial(F_\theta r \sin \theta)}{\partial \theta} + \frac{\partial(F_\phi r)}{\partial \phi} \right]$$

Since $F_r = r, F_\theta = 0, F_\phi = 0$,

$$\nabla \cdot \vec{F} = \frac{1}{r^2 \sin \theta} \frac{\partial(r^3 \sin \theta)}{\partial r} = \frac{3r^2 \sin \theta}{r^2 \sin \theta} = 3$$

Curl

For $\vec{F} = \hat{e}_1 F_1 + \hat{e}_2 F_2 + \hat{e}_3 F_3$,

$$\begin{aligned}\nabla \times \vec{F} = & \hat{e}_1 \frac{1}{h_2 h_3} \left[\frac{\partial(h_3 F_3)}{\partial q_2} - \frac{\partial(h_2 F_2)}{\partial q_3} \right] \\ & + \hat{e}_2 \frac{1}{h_1 h_3} \left[\frac{\partial(h_1 F_1)}{\partial q_3} - \frac{\partial(h_3 F_3)}{\partial q_1} \right] \\ & + \hat{e}_3 \frac{1}{h_1 h_2} \left[\frac{\partial(h_2 F_2)}{\partial q_1} - \frac{\partial(h_1 F_1)}{\partial q_2} \right]\end{aligned}$$

Curl in Spherical Coordinates

For $\vec{F} = \hat{e}_r F_r + \hat{e}_\theta F_\theta + \hat{e}_\phi F_\phi$,

$$\begin{aligned}\nabla \times \vec{F} = & \hat{e}_r \frac{1}{r^2 \sin \theta} \left[\frac{\partial(r \sin \theta F_\phi)}{\partial \theta} - \frac{\partial(r F_\theta)}{\partial \phi} \right] \\ & + \hat{e}_\theta \frac{1}{r \sin \theta} \left[\frac{\partial F_r}{\partial \phi} - \frac{\partial(r \sin \theta F_\phi)}{\partial r} \right] \\ & + \hat{e}_\phi \frac{1}{r} \left[\frac{\partial(r F_\theta)}{\partial r} - \frac{\partial F_r}{\partial \theta} \right]\end{aligned}$$

Example 8.24

Find the curl of the function $\vec{F} = \hat{e}_r \sin \phi$.

Using the previous formula,

$$\begin{aligned} \nabla \times \vec{F} = & \hat{e}_r \frac{1}{r^2 \sin \theta} \left[\frac{\partial(r \sin \theta F_\phi)}{\partial \theta} - \frac{\partial(r F_\theta)}{\partial \phi} \right] + \hat{e}_\theta \frac{1}{r \sin \theta} \left[\frac{\partial F_r}{\partial \phi} - \frac{\partial(r \sin \theta F_\phi)}{\partial r} \right] \\ & + \hat{e}_\phi \frac{1}{r} \left[\frac{\partial(r F_\theta)}{\partial r} - \frac{\partial F_r}{\partial \theta} \right] \end{aligned}$$

Since $F_r = \sin \phi$, $F_\theta = 0$, $F_\phi = 0$,

$$\nabla \times \vec{F} = \hat{e}_\theta \frac{1}{r \sin \theta} \frac{\partial(\sin \phi)}{\partial \phi} + \hat{e}_\phi \frac{1}{r} \left[-\frac{\partial(\sin \phi)}{\partial \theta} \right] = \hat{e}_\theta \frac{\cos \phi}{r \sin \theta}$$

Laplacian

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_1 h_3}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

In spherical polar coordinates,

$$\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}$$

9. Integral Calculus with Several Independent Variables

(9장 다변수 적분)

Single-Variable Integration

For a function $f(x) = dF/dx$,

$$dF = f(x)dx$$

$$\int_{x_0}^{x_1} dF = \int_{x_0}^{x_1} f(x) dx$$

$$F(x_1) - F(x_0) = \int_{x_0}^{x_1} f(x) dx$$

This is the integral **along the x axis** from x_0 to x_1

Line Integral (선적분)

Consider a differential with two independent variables:

$$dF = M(x, y)dx + N(x, y)dy$$

For the curve C joining (x_0, y_0) and (x_1, y_1) ,
the **line integral** (path integral) is defined as

$$\int_C dF = \int_C [M(x, y)dx + N(x, y)dy]$$

Evaluation of Line Integral

$$\int_C dF = \int_C [M(x, y)dx + N(x, y)dy]$$

The curve can be expressed as $x = x(y)$ or $y = y(x)$, so

$$\int_C dF = \int_C [M\{x, y(x)\}dx + N\{x(y), y\}dy]$$

$$\int_C dF = \int_{x_0}^{x_1} M\{x, y(x)\} dx + \int_{y_0}^{y_1} N\{x(y), y\} dy$$

Example 9.1

Find the value of the line integral

$$\int_C dF = \int_C [(2x + 3y)dx + (3x + 4y)dy]$$

where C is the straight-line segment given by $y = 2x + 3$ from $(0,3)$ to $(2,7)$.

Rewrite y in the first term in terms of x :

$$2x + 3y = 2x + 3(2x + 3) = 8x + 9$$

The integral becomes

$$\int_C (2x + 3y)dx = \int_0^2 (8x + 9) dx = (4x^2 + 9x) \Big|_0^2 = 34$$

Example 9.1

Find the value of the line integral

$$\int_C dF = \int_C [(2x + 3y)dx + (3x + 4y)dy]$$

where C is the straight-line segment given by $y = 2x + 3$ from $(0,3)$ to $(2,7)$.

Similarly, rewrite x in the second term in terms of y :

$$3x + 4y = 3(y - 3)/2 + 4y = (11y - 9)/2$$

The integral becomes

$$\int_C (3x + 4y)dy = \int_3^7 \left(\frac{11}{2}y - \frac{9}{2} \right) dy = \left(\frac{11}{4}y^2 - \frac{9}{2}y \right) \Big|_3^7 = 92$$



Hence, the total value of the line integral is $34 + 92 = 126$.